programmed rules and heuristics. They respond to sensor inputs according to fixed logic and a bounded set of conditions but do not learn, reason, or adapt their behavior based on new or unforeseen experiences outside of their defined envelope. Furthermore, these traditional OT systems typically operate in isolation, use proprietary protocols, and are focused primarily on real-time control and maintaining stability within set parameters, rather than holistic, future-oriented optimization of the entire asset or enterprise. The autonomous digital twin, in contrast, integrates agentic AI to reason about multiple, conflicting objectives, predict future states, and generate novel control strategies that adapt over time.

REAL WORLD EXAMPLE: MAINTAINING YOUR HOME TEMPERATURE

You don’t need to walk through an industrial plant to see an autonomous digital twin in action. In fact, you might already live with one—and it’s probably hanging on your wall.

Let’s say you’ve installed a smart thermostat in your home. When you first turn it on, it behaves like any other thermostat, where you configure the temperature that you want, and it turns on the heat or the air conditioning to bring the temperature as close to that as possible, very much like a traditional closed loop process control system.

But after running for a few days, things start to change. It learns that you tend to lower the temperature at around 10 p.m. when you go to bed, and then turn it up again at 7 a.m. when you wake up in the morning. It also reads the local weather data and starts to form a correlation as to how the external conditions change your home’s climate. It even communicates with your mobile phone to determine your physical location and uses this to infer when you are not at home. Eventually, it starts to make changes to the temperature settings on its own as shown in figure 1.11.